

Conformity review, Deliverable 3

Structural Aliasing in Patch-Based Time-Series Foundation Models — all coursework points, sources as of 26 August 2026

Verdict at a glance

#	Requirement	Verdict	What decides it
1	Tokenisation operator, patch identity	PARTIAL	Operator and the exact-identity condition are derived (<code>one.tex</code> , <code>two.tex:148–208</code>); <i>indistinguishable</i> is never formalised, and only sufficiency is proved.
2	f_s , P , S , f ; lock class; degeneracies	MET	Full derivation to $\mathcal{F}_{\text{lock}}$, <code>two.tex:16–131</code> , plus the three overlap regimes. Two gaps below.
3	Semantic layer and its exposure	PARTIAL	Ontology is substantial (Appendix A) but the <i>effect on concept recognition</i> is three sentences, and no forecast concept exists.
4	Falsifiable hypotheses (i)(ii)(iii)	PARTIAL	All three clauses covered with thresholds; the representational half of $H1$ has no pre-registered rule, and $M1$ ’s rule is mis-oriented.
5	Controlled experiments, isolation	MET	Isolation argued four independent ways; fifteen geometries verified arithmetically. Phase design undocumented.
6	Behavioural and representational	MET	Both levels genuinely present and contrasted; verdicts rest on fits the report itself disqualifies.
7	Bayesian analysis	PARTIAL	Priors, intervals and sensitivity are strong; the promised model comparisons are not reported, and an undisclosed filter conditions on the outcome.
8	Mitigation strategy	MET	$M1$ pre-registered with a principled geometric explanation (<code>eight.tex:11</code>); the numerical verdict is unsafe.
9	Agentic AI	MET	Represent / monitor / explain / act all present, <code>eight.tex:15–30</code> .
10	NIST AI RMF risk assessment	PARTIAL	Threats, surfaces, consequences and residual exposure all present; the RMF functions are a table column, not structure.

Word count. Body text 7,180 + headings 155 = **7,335 words** against the 8,000 + 10% = 8,800 limit. Captions (135), tables, figures, references, appendices and the executive summary are excluded. Roughly 1,400 words of headroom remain. The executive summary is 389 words, within its 400 cap — but it is still Lorem ipsum.

Blockers: fix these before submission

- B1. The executive summary is Lorem ipsum.** `sections/summary.tex:1`, rendered on page 1 of `main.pdf` under “Executive Summary”. This is the first thing the marker reads.
- B2. “CHATGPT ACADEMIC PROPOSAL” is in the body.** `sections/eight.tex:21`, a bare heading followed by 22–24, which are a paraphrased duplicate of 17–19. It renders as body text inside the section carrying *both* requirement 8 and requirement 9. Delete lines 21–24.
- B3. Stray `[]` typeset in Appendix A.** `main.tex:139–140`: the `[]` after each `\let` is not an argument and is printed. Remove both.
- B4. The δ_O sign is inconsistent, and it inverts the $M1$ verdict.** Model A is fitted on the contrast d (`four.tex:24–40`), where *more negative is worse*; the appendix derives from this that a mitigating overlap must *lift* β_c , i.e. $\delta_O > 0$ (`bayesian_appendix.tex:130`). The pre-registered rule, however, demands $\Pr(\delta_O < 0) \geq 0.95$ (`tables/bayesDecisions.tex:13`, `four.tex:107`), which is the rule for the flipped deficit $y = -d$ used by Model C. `four.tex:96` and `four.tex:107` contradict each other eleven lines apart. On the orientation the model is actually fitted on, the reported posterior gives $\Pr(\delta_O > 0) = 0.964$ and 0.977 — the mitigation direction. `eight.tex:7` declares $M1$ **refuted**; that is the exact inversion of the specification. The error propagates to `eight.tex:19,34`, `ten.tex:33,35` and risk item T4 including its “high” residual rating. *This is the most serious single defect in the report.*
- B5. The convergence gate is stated four times and broken three times.** `six.tex:27` (“Only a posterior that passes those checks is carried into the three-way rule”), `four.tex:116`, `bayesian_appendix.tex:7`,

`seven.tex:15`. Yet Model B (ESS 676/525, below the 1000 threshold) is declared *supported* at `seven.tex:30`; Model C ($\hat{R} > 1.6$) is declared *supported* at `seven.tex:37` with the gate overridden in prose; Model A ($\hat{R} > 1.3$, ESS < 12) issues *refuted* at `eight.tex:7`. Either soften the rule to permit qualified verdicts, or withdraw the three labels.

- B6. An undisclosed, outcome-dependent row filter.** `chronos/bayesian/collect.py:166-168` sets `live = max(R_lock, R_lo, R_hi) > 0.05` and every Model A and Model C fit, the recovery simulation and the sensitivity ladder run on `live` rows only. The report never mentions the filter, the threshold, or any row-level exclusion — `six.tex:11` discloses only the control-pair and stride-boundary rules. The filter conditions on the response: it removes exactly the cases where the model rebuilt nothing, which is the outcome *H1* predicts, so it biases *H1*-behavioural toward the null that was reported. Disclose and justify it, or drop it.
- B7. The three Appendix E pages are still placeholders.** `figures/` holds only the four H1–H3 PNGs, so `sweep_appendix.tex:63,80,95` fire their `\IfFileExists` fallbacks and `main.pdf` prints three framed boxes reading “FIG_E1_pageN.png not yet copied”. Meanwhile `seven.tex:5` claims both readings “are reproduced” there, and `seven.tex:7`’s sweep numbers (12–27% recovery, 127–133 Hz RMSE, probe 0.77 vs 0.85) have no figure or table backing them anywhere. Copy the three PNGs in, or remove the dependent claims.
- B8. A frequentist analysis is claimed and does not exist.** `seven.tex:70`: “Where Bayesian and frequentist analyses address the same question, they agree”. No frequentist result, method, *p*-value or bootstrap appears anywhere in the document. On a brief that explicitly contrasts Bayesian inference with NHST this is a costly sentence to be caught on. Delete it.

Findings by requirement

1 — Tokenisation operator

PARTIAL

Present: $\mathcal{T}_{P,S}$ with domain and codomain, `one.tex:1-28` (`eq:token_def`, `eq:element_def`); consecutive-patch identity `two.tex:148-170`; the $j = P/\gcd(P, S)$ period `two.tex:174-208`; reconstruction and injectivity for $S \leq P$ `two.tex:212-225`.

- “Indistinguishable” is never formalised. Only exact equality is derived — no ε -neighbourhood, no bound relating a detuning Δf to $\|t_{k+1} - t_k\|$. Yet `ontology.tex:132-140` depends on tolerances $\varepsilon_S, \varepsilon_P$ that are asserted rather than derived, and `six.tex:11` picks $\delta \leq 0.25f_s/S$ by fiat.
- *Sufficiency only.* `eq:general_periodicity` assumes *global* invariance $x[n+S] = x[n] \forall n$; the exact condition for $t_{k+1} = t_k$ is the local one, over $m = 0, \dots, P-1$. For the non-sinusoidal signals `two.tex:133` claims to cover, the gap is real.
- *The third clause is answered by refutation without saying so.* `two.tex:225` concludes that no sub-Nyquist frequency is destroyed by tokenisation — correct, and the honest reframing at `two.tex:258-280` is a strength — but “loss of observability” is never defined at the signal/token level, while `eight.tex:13,28` and T2 use the phrase without the representation-level qualifier that `two.tex` says is mandatory.
- Requirement 1’s derivations live in §2, not §1, and no requirement-to-section map exists. A marker reading §1 as the answer sees 28 lines of definitions. *Consider one sentence at the head of §1 pointing forward.*

2 — Lock class and degeneracies

MET

The derivation is complete and, checked step by step, correct: `two.tex:16-131` through to $\mathcal{F}_S = \{cf_s/S\}$, $\mathcal{F}_P = \{kf_s/P\}$ and `eq:flock_set`, with the sampling-rate-free normalisation `eq:cpp`. Two gaps worth a sentence each:

- *The token-space degeneracy class is larger than $\mathcal{F}_{\text{lock}}$.* $t_{k+j} = t_k$ holds whenever $jSf/f_s \in \mathbb{Z}$, i.e. at every rational cps. Inside the report’s own design: $P = S = 16$, $f = f_s/32$ gives $t_{k+2} = t_k$ at a frequency `eq:flock_set` excludes. The $j > 1$ case is derived for the patch branch and stopped at $j = 1$ for the stride branch.
- *Existence condition unstated:* $c < S/2$ and $k < P/2$ for a locked frequency below Nyquist.
- **Overclaim at `two.tex:260-262`:** “the sequence repeats every $P/\gcd(P, S)$ tokens while neighbours stay distinct unless $S = P$ ”. False on $\mathcal{F}_S \cap \mathcal{F}_P$ — e.g. $P = 16, S = 8, f = f_s/8$, where neighbours are identical with $S \neq P$. It also contradicts the caption of `tab:hfModels`. The correct statement is made properly at `two.tex:174`; the summary sentence generalises it wrongly.
- `tables/flockDomains.tex` and `tables/patchStride.tex` exist on disk but are never `\input`, so $\mathcal{F}_{\text{lock}}$ is never instantiated numerically in the document.

3 — Semantic layer

PARTIAL

Concepts (17 classes), states (five closed vocabularies), relations (15 object properties), routing matrix and decision tree are all present and formalised, `ontology.tex:1-277`. The second clause is the weak one.

- *The effect on concept recognition is three sentences* (`ontology.tex:186`, `three.tex:3-5`). No worked case of a `RampEvent` missed or spuriously asserted, no analysis of which routes could be mis-selected. The only

quantitative link (`eight.tex:11`) sits in §9.

- **No forecast concept exists in the ontology.** No `Forecast`, `Prediction`, `ForecastModel` or horizon property in any of the four tables; every rule consumes *measured* values, and `eq:semantic-path` has no forecast node. So the claim at `ontology.tex:186` that aliasing reduces confidence in “forecast-derived production, ramp expectations, forecast-driven anomaly recognition” has no referent in the formal representation. *Adding one class and one relation would close this.*
- **Undefined symbol:** `eq:locking-candidate` defines C ; the prose calls the union Z ; `eq:observability-state` uses $X = Z \wedge Y$. Z is never defined.
- **OWL2 overclaim:** `introduction.tex:21` says the layer “is formalised as an OWL2 ontology”; what is delivered is prose plus four tables, with no axiom listing or serialisation, and `ontology.tex:65` concedes the route constraints are pushed to an external validator.
- **Threshold mismatch:** the semantic layer records $\tau > 1/2$ while the analysis applies 0.95/0.05.

4 — Falsifiable hypotheses

PARTIAL

(i), (ii) and (iii) are each covered, with estimands, models and numeric thresholds in `tab:bayesModels`, `tab:bayesPriors` and `tab:bayesDecisions`, and the three-way convention at `four.tex:112`.

- **$H1$ ’s representational half has no pre-registered rule.** `four.tex:51` gives a direction only; `tab:bayesDecisions` has no row for θ_{lock} . `seven.tex:30` then decides it against $\Pr > \log 1.2 = 1.00$, a threshold appearing nowhere in the table — which falsifies the claim at `four.tex:112` that every threshold was fixed before a posterior was inspected. This matters because the representational result is the report’s headline finding.
- $M1$ ’s rule is mis-oriented (B4).
- $H3$ -location’s rule needs $\theta_P < 0$, and θ_P is never reported.
- $H2$ is confounded with $H1$ -behavioural and this is not acknowledged. σ_φ is the spread of per-phase offsets on the same response where `seven.tex:20` finds no deficit at all. With no deficit to modulate, $\sigma_\varphi \approx 0$ is near-guaranteed, so “ $H2$ supported” may be an artefact of the behavioural null. Phase enters no representational model. One caveat sentence in §8 would defuse this.
- **Design claim contradicted by result:** `four.tex:92` says `tab:hfModels` supplies both series; `seven.tex:57` reports $H3b$ “not identified”.

5 — Controlled experiments

MET

Isolation from classical sampling is argued four independent ways — band strictly inside Nyquist (`six.tex:9`), the operator proved lossless for $S \leq P$ (`two.tex:214-225`), the phenomenon defined as representation-level (`two.tex:258-280`), and matched triplets sharing background and phase (`four.tex:22`, `six.tex:11`). `tab:hfModels` was verified arithmetically: every overlap ratio and token count is right, exactly six runs have $S \nmid P$, and $\text{corr}(\hat{O}, \log \hat{P}) = 0.425$ matches the 0.43 quoted in the appendix.

- *The phase design is never specified.* `six.tex:11` says only “the non-redundant alignments available at each frequency” — the number is unstated and by that wording varies with frequency, so the phase axis is unbalanced and untabulated. Of the two axes the brief names, patch parameters get a full table and temporal offsets get one clause.
- *The “full set of twenty-two” is never listed* (`seven.tex:1`); roughly half the numbers in §8 come from a design the report does not describe. The released code is also inconsistent with the count: `probe_lib.py` defines 22 then removes $S \in \{5, 15, 28\}$, leaving 17, and `collect.py` fits those 17. $480 \bmod 5 = 480 \bmod 15 = 0$, so this is not the stride-boundary rule.
- **`seven.tex:5` and `sweep_appendix.tex:1` describe different experiments:** five geometries versus fifteen, under one cross-reference.
- **`six.tex:17` claims results the appendix says were not produced:** “the pure, TSMixup and KernelSynth views show corresponding frequency holes” — `sweep_appendix.tex:11` says the sweep is pure sinusoids only, and the TSMixup/KernelSynth comparison sits in three “Planned extension” subsections.
- *Broken sentence at `six.tex:3`:* a subordinate clause with no main clause.

6 — Behavioural and representational evaluation

MET

Both levels are present and, unusually, explicitly contrasted rather than merged — the “representation penalised, forecast intact” reading at `seven.tex:32-33` is the report’s best result. Behavioural: $R = A_{\text{pred}}/A_{\text{true}}$ in matched triplets, `fig:forest`. Representational: MDL probing over ten captured states per geometry (`bayesian_appendix.tex:49`) and pre-block token-embedding dispersion (`bayesian_appendix.tex:83`), `fig:combs`.

- Spectral *reconstruction* in the brief’s sense (input spectrum vs output spectrum) is planned, not executed — three “Planned extension” subsections plus a T0 D0 comment block left in `sweep_appendix.tex:154-158`. *Remove the TODO comment at minimum.*
- θ_P is never reported and no verdict is stated for $H3$ -location at all.
- **No LOO numerics appear anywhere.** `bayesian_appendix.tex:9` defines the reading rule (`elpd_diff <`

2dse is inconclusive) and five comparisons are promised; only a bare “wins LOO” survives (`seven.tex:48`). The second leg of the *M1* rule is therefore never evaluated, so *M1* cannot be adjudicated in *either* orientation.

- `fig:phasePosterior` — the sole figure evidencing requirement 4(ii) — is never `\autoref`’d; nor are `fig:sweepPage2` and `fig:sweepPage3`.
- *Divisibility slip* at `seven.tex:57`: the patch branch is said to be unidentified because “in most configurations $S \mid P$ ”. $S \mid P$ removes stride-only sites; what removes patch-only sites is $P \mid S$, which holds in only 4 of 15 runs.

7 — Bayesian analysis

PARTIAL

Priors are stated *and* justified individually, with prior-predictive baselines per decision rule — above the expected standard. Uncertainty is quantified throughout with intervals, ROPE probabilities, a parameter-recovery check and a prior/likelihood sensitivity ladder.

- **Model comparison is promised four times and reported once, by rank alone.** `six.tex:23` promises LOO for Models A, B and C; only D1’s appears, and without `elpd_diff` or `dse`, in violation of the report’s own reading rule. This is the clause of requirement 7 the brief asks for most directly. *Report the numbers or state plainly that the comparisons were not run.*
- The convergence-gate contradiction (B5) and the undisclosed `live` filter (B6) both land here.
- *Selective set reporting*: `ten.tex:17` and T2 present the representational cost as “confirmed” on the clean-set figure, while `seven.tex:30` records the full set as inconclusive.

8 — Mitigation

MET

M1 is pre-registered with a decision rule and, at `eight.tex:11`, given a geometric explanation that stands independently of the fit: shortening *S* raises f_s/S and thins the stride branch, but leaves f_s/P invariant, so overlap cannot touch the patch branch — instantiated on the one-minute PV telemetry. Two competing readings of the positive slope are offered. This is the strongest passage in the report and it survives everything else.

- `eight.tex:3` states only the sign leg of the rule; `bayesian_appendix.tex:134` and `tab:bayesDecisions` require *both* legs and say “Both legs must hold”.
- The numerical verdict is drawn from a fit with $ESS < 12$ (B5), on an orientation that inverts it (B4). The geometric argument survives; the number does not.

9 — Agentic AI

MET

All four verbs are present and grounded in the ontology: represent (`eight.tex:28`), monitor (`eight.tex:28`, with the load-bearing point that the detector must sit *upstream* of tokenisation), explain (`eight.tex:30`, assessments carrying estimand, effect size, interval and posterior probability), act (`eight.tex:19,30`, bounded reversible degradation of authority).

- “Human-facing explanations” is the thinnest clause — one sentence, no worked artefact or model-card entry shown.
- `ontology.tex:151` requires a pre-declared criterion before an instance may be asserted `StructurallyAliased`; on this report’s own evidence that trigger is never reached, and `eight.tex:19` describes what a confirmed assessment does without noting that none was produced.

10 — NIST AI RMF

PARTIAL

Threat models (T1 adversarial masking with a five-element model; T2–T5), attack surfaces (telemetry upstream of tokenisation, reached by driving a periodic load rather than through the data path), consequences in all three named contexts, and an unusually honest residual-exposure statement (`ten.tex:29–35`) are all present. `ten.tex:19–21` explicitly withdraws the strong form of T1 as unsupported, which reads well.

- **The RMF functions are labels, not structure.** GOVERN/MAP/MEASURE/MANAGE appear only as one column of `tab:riskRegister`. No prose defines them, no category or subcategory is cited (e.g. MAP 1.1, MEASURE 2.7), and GOVERN is never discussed at all. This reads as a risk register with RMF tags applied afterwards rather than an assessment organised by the Framework — *the single clause most likely to be marked down on the Computer Security half*. Two short paragraphs organising §10 under the four functions, with subcategory citations, would fix it.
- Two of the three “strengthening” findings rest on models the report’s own gate excludes, and neither caveat is carried into §10 or the register.

Smaller items

- Dead files never `\input:` `sections/three_proposal.tex`, `tables/flockDomains.tex`, `tables/patchStride.tex`, `tables/trainParams.tex`, `tables/tsmixupPool.tex`. `sections/nine.tex` is an empty stub (merged into `eight.tex`).
- Symbol collision: k is the token index and the patch-harmonic integer in the same section, and is reused twice more in the ontology. `cps/cpp` are silently renamed ρ_S, ρ_P in `ontology.tex:126-128`.
- `seven.tex:1` claims qualitative conclusions hold on both sets; `seven.tex:30` reports a change of verdict between them.
- `seven.tex:15` gives D1/D2 as $\text{ESS} > 1000$, `seven.tex:48` as $\text{ESS} > 5000$.
- `main.log` predates `main.pdf`, so it cannot certify a warning-free build. Recompile before submitting and check for undefined references.

Suggested order of work

1. Write the executive summary; delete `eight.tex:21-24`; delete the two `[]` in `main.tex`; delete `seven.tex:70`. *Half an hour, removes four unforced errors.*
2. Settle the δ_O orientation and rewrite the $M1$ verdict everywhere it propagates. Expect *inconclusive*, not *refuted*: the clean-set interval crosses zero and the LOO leg was never evaluated.
3. Reconcile the convergence gate with the three verdicts that break it — one qualifying sentence in §8 is enough, but it has to be there.
4. Disclose the `live` filter.
5. Copy the three `FIG_E1_page*.png` into `figures/`, or drop the claims that depend on them.
6. Add a θ_{lock} row to `tab:bayesDecisions`, report θ_P and the LOO numbers, or say they were not run.
7. Give §10 two paragraphs organised by the four RMF functions, GOVERN included.